

CMPG 325 - Milestone 2

Client Implementation Review

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Contents

1	Working Packet Tracer File	2
2	Assigned Feature Implemented: Default Routing (Edge/ISP Path Design)	2
3	Testing Evidence	3
3.1	Baseline connectivity	3
3.2	Internet reachability and NAT	4
3.3	Default Routing failover — the assigned challenge’s core evidence	5
3.4	Reception access restriction	7
4	Troubleshooting Summary	9
A	Device Configuration Summary	10

1 Working Packet Tracer File

The network was built to the design finalised at Milestone 1: 192.168.41.0/24 split across five department VLANs (Management, Employment Services, Finance, IT/Servers, Reception), with R-EDGE as the edge router performing inter-VLAN routing, DHCP relay, NAT, and dual-ISP default routing.

Hardware note: R-EDGE is a 4321/4331 router. The LAN trunk to SW-CORE runs on the onboard GigabitEthernet0/0/0 interface. The Backup ISP link required an added **NIM-2T serial module** (Serial0/1/0) after an initially-tried Ethernet switch module (NIM-ES2-4) proved unable to support routed IP addressing — see Section 4 for the full troubleshooting account.

Full device configurations are included in `configs/` and summarised in Appendix A.

File: taung-labour-network.pkt

2 Assigned Feature Implemented: Default Routing (Edge/ISP Path Design)

What was configured:

```
ip route 0.0.0.0 0.0.0.0 192.168.41.146 1 ! Primary default route (AD 1)
ip route 0.0.0.0 0.0.0.0 192.168.41.150 5 ! Floating default route via Backup ISP (AD5)
```

Why this is appropriate: a floating static default route is the standard, low-complexity mechanism for automatic ISP failover without introducing a dynamic routing protocol, matching the challenge’s “Intermediate” difficulty rating. Because the administrative distance of the backup route (5) is higher than the primary’s (1), the router only installs the backup route into its active routing table when the primary route is withdrawn — giving automatic, self-healing failover with no manual intervention.

How this satisfies the design constraint: the brief requires a backup internet path specifically for the Finance function. Rather than building Finance a separate physical path, the whole network shares one edge router with this failover behaviour, so Finance’s internet access survives a primary-link failure along with every other department — a simpler design that still fully satisfies the stated requirement.

Dual NAT overload: initial testing (Section 4) revealed that a single NAT overload rule tied to the primary interface left the network without a working internet translation path whenever the primary was down — internally correct routing, but no usable internet access. This was resolved by adding a second NAT overload rule for the backup interface, so translation works correctly regardless of which path is active:

```
ip nat inside source list NAT_ACL interface GigabitEthernet0/0/1 overload
ip nat inside source list NAT_ACL interface Serial0/1/0 overload
```

3 Testing Evidence

3.1 Baseline connectivity

show vlan brief (SW-CORE) — confirms all six VLANs (10, 20, 30, 40, 50, 99) are active.

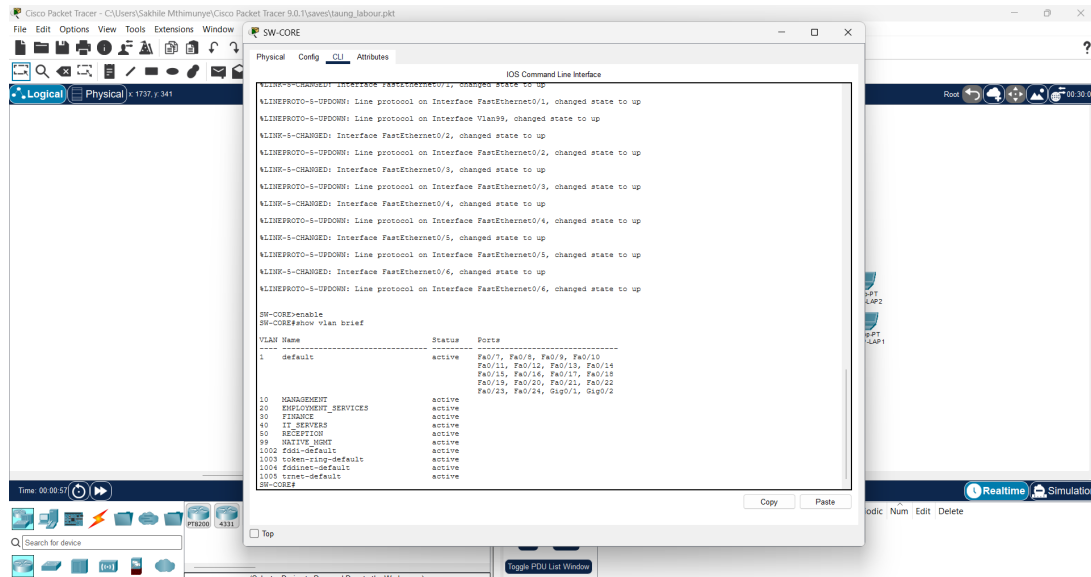


Figure 1: vlan brief

show interfaces trunk (SW-CORE) — confirms all six ports (Fa0/1–Fa0/6) are trunking, native VLAN 99, with VLANs 10, 20, 30, 40, 50, 99 allowed on every port.

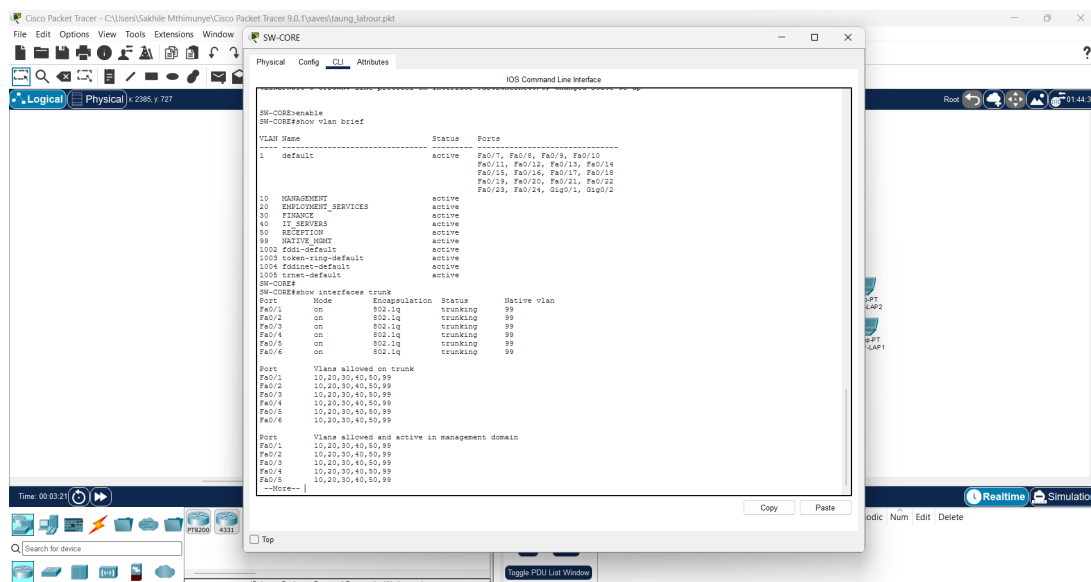


Figure 2: interfaces trunk

show ip route (R-EDGE), primary path active — confirms every VLAN subnet is directly connected and the gateway of last resort is the primary ISP (192.168.41.146), with administrative distance 1.

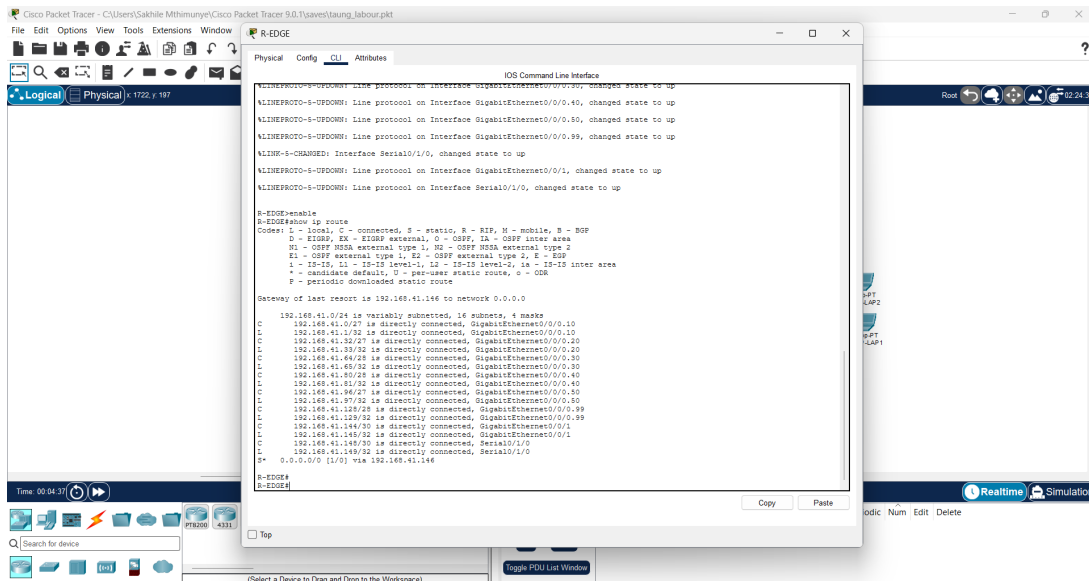


Figure 3: ip route

3.2 Internet reachability and NAT

ping 8.8.8.8 succeeding from a PC in each VLAN:

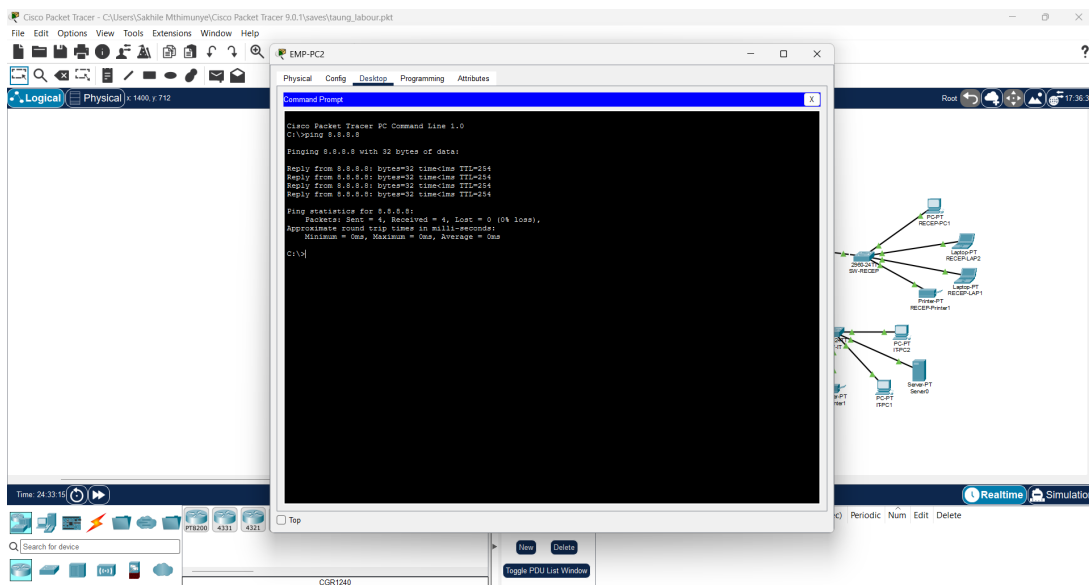


Figure 4: successful ping 8.8.8.8

show ip nat translations showing a live entry after a successful ping

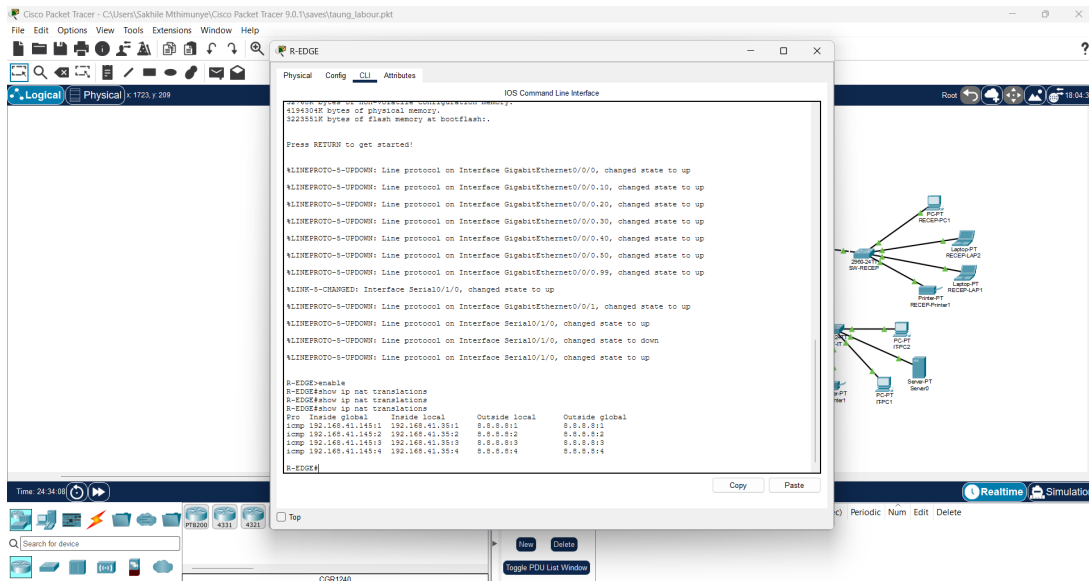


Figure 5: ip nat translations

3.3 Default Routing failover — the assigned challenge’s core evidence

show ip route before the outage (gateway of last resort via 192.168.41.146):

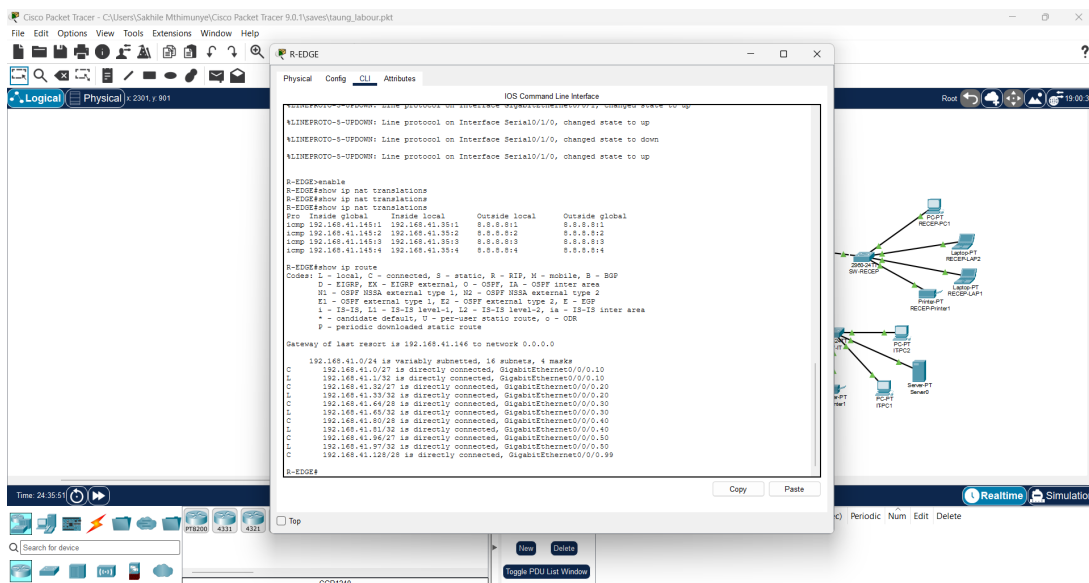


Figure 6: ip route before

show ip route immediately after shutdown on GigabitEthernet0/0/1 (gateway of last resort switches to 192.168.41.150 via Serial0/1/0):

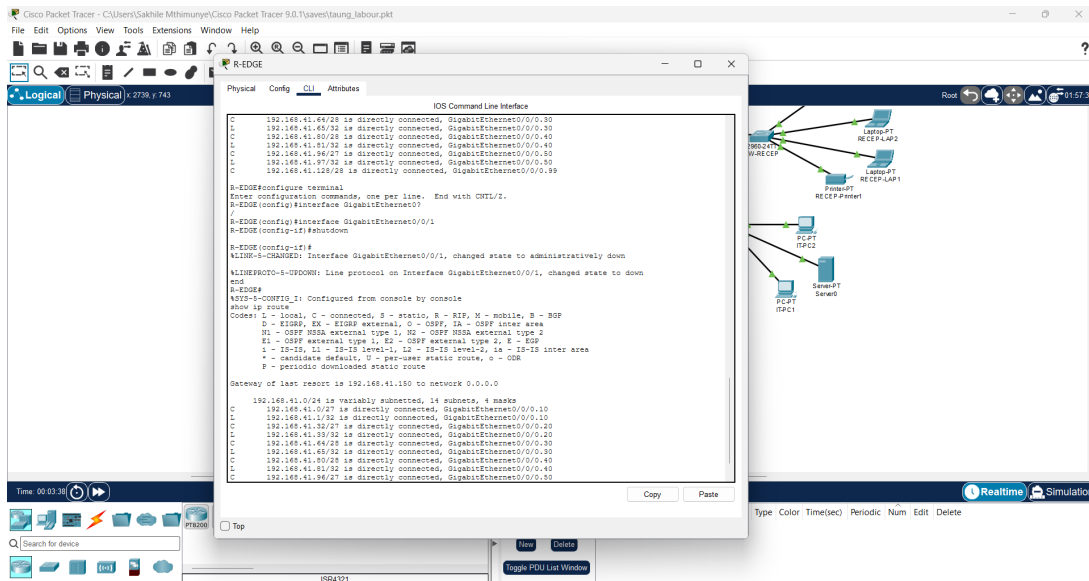


Figure 7: ip route after shutdown

ping 8.8.8.8 from a **Finance** PC succeeding while the primary interface is down:

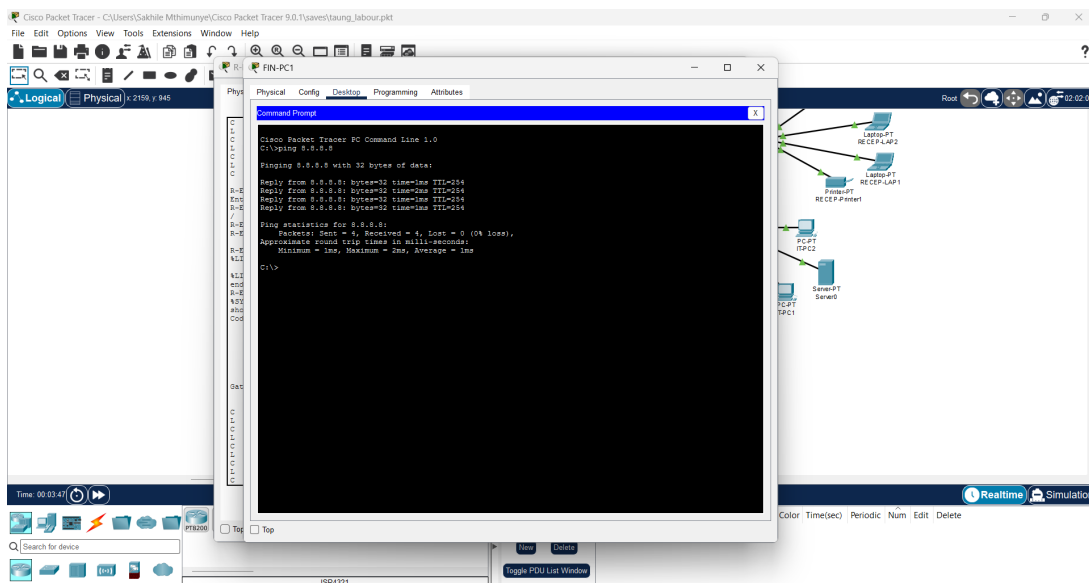
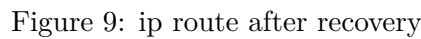


Figure 8: ping 8.8.8.8 from finance pc

show ip route after no shutdown on GigabitEthernet0/0/1, confirming the primary path is preferred again once restored:



ping from a Reception PC to a Finance or IT host — request times out (blocked by `RESTRICT_RECEPTION`):



The same ping from a Management or Employment Services PC — succeeds (unaffected departments):

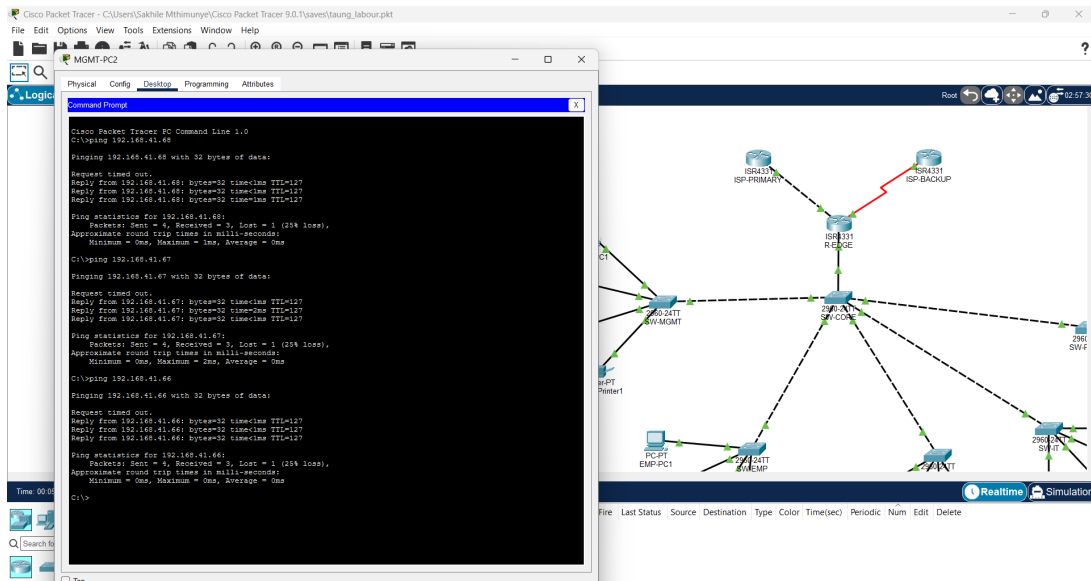


Figure 11: ping from management to finance allowed

ping `intranet.taunglabour.gov.za` configured DNS record from a Reception PC — the first output line shows the name correctly resolved to 192.168.41.85 even though the subsequent ICMP replies destination host unreachable, confirming the ACL's DNS permit rule works independently of the ICMP block:

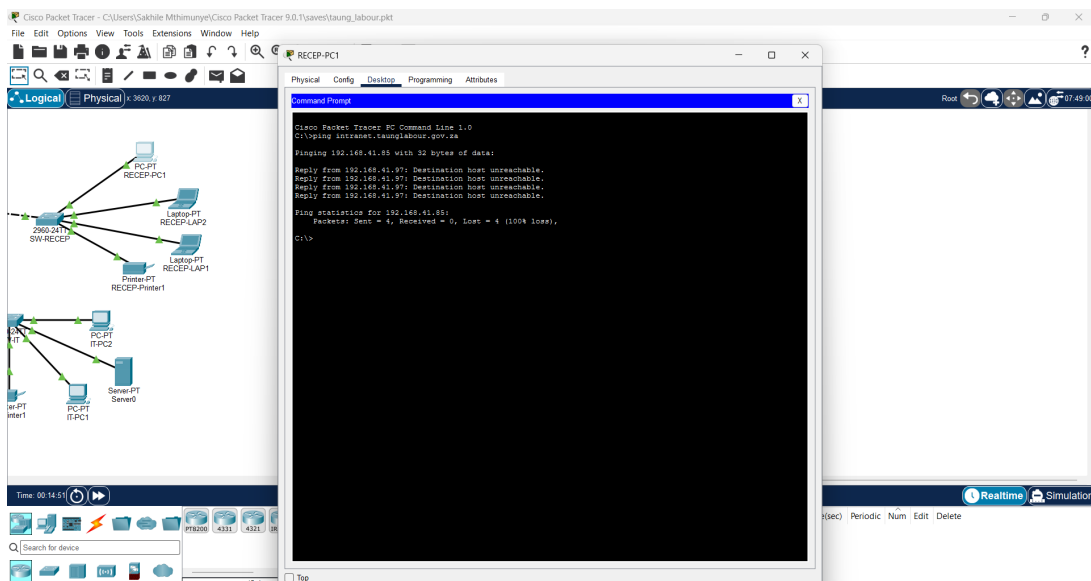


Figure 12: Reception DNS still works

4 Troubleshooting Summary

Four issues were identified and resolved during implementation:

1. **Config commands rejected on switches** — commands were being entered at user EXEC mode (`Switch>`) rather than global config mode. Resolved by always running `enable` then `configure terminal` first.
2. **Router's third interface could not support VLAN sub-interfaces** — the NIM-ES2-4 module added to reach three physical interfaces is a Layer-2 switch module, not a routed interface, and rejected `encapsulation dot1Q` sub-interface creation even after attempting `no switchport`. Resolved by replacing it with a NIM-2T serial module, moving the Backup ISP link to `Serial0/1/0` and keeping the LAN trunk on the onboard, sub-interface-capable `Gi0/0/0`.
3. **Primary ISP link showing up/down mismatch** — `show ip interface brief` on both ends revealed the IP configuration had been applied to `Gi0/0/1`, while the physical cable was connected to `Gi0/0/0`. Resolved by matching the configuration to the interface that was actually cabled.
4. **Internet unreachable despite correct-looking configuration** — diagnosed in two parts: first, `8.8.8.8` did not exist anywhere in the simulated topology (resolved by adding loopback `8.8.8.8/32` interfaces to both ISP devices as simulated internet hosts); second, `show ip nat translations` remained empty while the network was running on the backup path, traced to the NAT overload rule only referencing the primary interface (resolved by adding a matching NAT overload rule for the backup interface, per Section 2).

Full narrative detail is maintained in `reflection.md`.

A Device Configuration Summary

Device	Role	Config file
R-EDGE	Edge router — WAN links, VLAN sub-interfaces, default routing, dual NAT, Reception ACL	<code>configs/R-EDGE.txt</code>
SW-CORE	Core switch — trunks to router and all access switches	<code>configs/SW-CORE.txt</code>
SW-MGMT	Access switch — VLAN 10 Management	<code>configs/SW-MGMT.txt</code>
SW-EMP	Access switch — VLAN 20 Employment Services	<code>configs/SW-EMP.txt</code>
SW-FIN	Access switch — VLAN 30 Finance	<code>configs/SW-FIN.txt</code>
SW-IT	Access switch — VLAN 40 IT/Servers (hosts Server0)	<code>configs/SW-IT.txt</code>
SW-RECEP	Access switch — VLAN 50 Reception	<code>configs/SW-RECEP.txt</code>

Full listings are in the `configs/` folder of the GitHub repository.